

EMILY Y. XU

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EDUCATION

California Institute of Technology (Caltech)

Expected Graduation: 06/2027

B.S., Electrical Engineering - Intelligent Systems **GPA: 4.2/4.0**

Pasadena, CA

Coursework: Large Language Models for Reasoning, Machine Learning/Data Mining, Deep Learning, Data Structures/Algorithms, Software Design, Digital Logic and Embedded Systems, Deterministic Analysis of Systems and Circuits, Mathematics and Physics of Electrical Engineering, Electronic Systems Lab, Probability and Statistics, Robotics. **Intended:** Digital Design with FPGAs, Mixed-Mode Integrated Circuits, Advanced Topics in Vision, Probability Models, Networks, Control Systems.

EXPERIENCE

Machine Learning/Software Engineering Intern

June 2025 – Present

hireEZ

Remote — Mountain View, CA

- Developing a full-stack job application portal using **React**, **FastAPI**, and **PostgreSQL**, with secure user authentication and account features powered by **FastAPI Users** and **Supabase**.
- Built an LLM-powered resume-job matching pipeline using **text embeddings**, **cosine similarity**, and the **OpenAI API**, integrating **Pydantic-AI** to autofill profiles from parsed resume data.

Machine Learning Research Intern

Feb 2025 – Present

Autonomous Robotics and Control Laboratory, Advised by Prof. Soon-Jo Chung and Dr. Thomas Berrueta

Pasadena, CA

- Developed and injected **realistic disturbance models** into a custom simulator; generated training data and processed telemetry to train **neural networks** on **residual dynamics** for **meta-learning**-based adaptive control in the **Indy Autonomous Challenge**.
- Built a **physics-based vehicle simulator** from throttle input to wheel dynamics, integrating **drivetrain**, **engine**, and **transmission** models with **PID-based lateral and longitudinal control**.
- Ported the **Python** simulation to **C++** for integration with real-time **racecar control software**, enabling **embedded deployment** and compatibility with closed-loop simulators.

Physics Research Intern

Jun 2024 – Aug 2024

Caltech Observational Cosmology, Advised by Dr. Sunil Goihwala and Brandon Sandoval

Pasadena, CA

- Designed aluminum capacitor fin structures for quasi-particle trapping in **Kinetic Inductance Detectors (KIDs)** using the **Sonnet RF simulator**. Applied microwave circuit theory and quality factor analysis to increase phonon collection efficiency.
- Developed Python algorithms using linear regression and least squares optimization to automate extraction of quality factors from simulation data. Delivered technical presentations to researchers at **Fermilab** and **NASA JPL**.

PROJECTS

Photodiode Array for Real-Time Hand Gesture Recognition 🦊

- Designed and implemented a custom 2D photodiode sensor array with trans-impedance amplifier (TIA) front-end circuitry for analog signal conditioning; created circuit schematics using Fritzing and validated signal fidelity and bandwidth with LTSpice simulations.
- Integrated the analog frontend with an Arduino-based data acquisition pipeline; processed real-time intensity matrices and trained a Python neural network for gesture classification.

Infinite Vertical Platformer Video Game 🦊 | C, SDL, POSIX API

- Developed a full-stack, single- and multiplayer 2D video game inspired by Doodle Jump from scratch in C, using SDL for graphics rendering and the POSIX API for system-level programming.
- Implemented a custom physics engine, enemy AI, keyboard controls, UI, and unit testing framework; optimized memory usage and algorithm performance for real-time gameplay.

LEADERSHIP

Mission Lead, Caltech Air and Outer Space

Oct 2023 – Nov 2024

National Aeronautics and Space Association (NASA)

Pasadena, CA

- Directed a team of 10 Caltech students in crafting the mission concept and background research review for a 26-page proposal.
- Secured \$150,000 in funding to advance research on the proposed inflation technology, performed fluids simulations in COMSOL.

Canadian Team Captain

Feb 2022 – Aug 2022

International Young Physicists' Tournament

Timisoara, Romania

- Led a team of 5 in a thesis-style competition solving 17 open-ended physics problems with experimental and theoretical analysis.
- Built a 3-axis Arduino system with MATLAB control and Excel logging; ran EM simulations in COMSOL and Python.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, SQL, ABEL (HDL), $\text{\LaTeX}$

Frameworks & Libraries: TensorFlow, Scikit-learn, PyTorch, Flask, FastAPI, React, Dash, Plotly, Matplotlib, OpenCV, Keras

Tools: MATLAB, PostgreSQL, Docker, Arduino, SolidWorks, COMSOL, Sonnet, DipTrace, Git, VS Code, LTSpice, Foxglove

Skills: Circuit Design, Machine Learning, Deep Learning, Data Analysis, Data Mining, Scientific Writing, Research Methodologies, OOP

AWARDS

International Young Physicists' Tournament: Bronze 2022

Canadian Young Physicists' Tournament: 2x Gold 2022, 2023

Carnegie Mellon University Women's Mathematics Competition: Overall Third Place 2022